Intervals of Convergence with Endpoint Checks

The ratio test, endpoints from the center and radius, substituting an endpoint, and deciding convergence there.

Each power series below is a model, and a model is only usable on the set of inputs where its series converges — that set is what you are finding. Work in this order every time:

- Ratio test on $|a_{n+1}/a_n|$. Factor the $|x-c|$ out; the number L left over gives the radius $R = 1/L$ (and $R = \infty$ when $L = 0$).
- Endpoints: $x = c - R$ and $x = c + R$. State them in the model's units.
- Substitute each endpoint *separately.* The ratio test is silent there, so name the test you use — p-series, alternating series, nth-term, or a comparison.
- Write the interval with the bracket each endpoint earned.

1. A pressure sensor's correction factor is modeled by

$$C(x) = \sum_{n=1}^{\infty} \frac{(x-4)^n}{n \cdot 2^n},$$

where x is the measured pressure in bars. Apply the ratio test to $|a_{n+1}/a_n|$ and factor out $|x-4|$. Compute the numerical limit L that remains, then state the radius of convergence R in bars.

Answer: ________________

2. The series in problem 1 is centered at $x = 4$ bars with radius $R = 2$ bars. Compute the two endpoints of its interval of convergence, in bars. (Use $c - R$ and $c + R$.)

Answer: ________________

3. Substitute the left endpoint $x = 2$ bars into the series of problem 1. Simplify the powers of 2 so the result is a numeric series in n alone, name the test that settles it, and — if it converges — give the exact sum.

Answer: ________________

4. Now substitute the right endpoint $x = 6$ bars into the series of problem 1. Simplify it to a numeric series in n alone, name the test that settles it, and state whether the endpoint belongs to the interval. Then write the complete interval of convergence in bars, using the bracket each endpoint earned.

Answer: ________________

5. A control loop's gain is modeled by

$$G(x) = \sum_{n=1}^{\infty} \frac{(x+3)^n}{n^2 \cdot 5^n},$$

where x is the input offset in volts. Apply the ratio test, factor out $|x+3|$, compute the numerical limit L that remains, and state the radius of convergence R in volts.

Answer: ________________

6. Test both endpoints of the gain model in problem 5. The series is centered at $x = -3$ volts with radius $R = 5$ volts, so the endpoints are $x = 2$ and $x = -8$ volts. For each endpoint, simplify to a numeric series in n alone, name the test that settles it, and give the exact sum. Then write the complete interval of convergence in volts.

Answer: ________________

7. A reaction-rate model is

$$R(x) = \sum_{n=0}^{\infty} \frac{(x-2)^n}{n!},$$

where x is the temperature in kelvins above a reference value. **(a)** Apply the ratio test and compute the numerical limit L that remains after factoring out $|x-2|$. **(b)** State the interval of convergence, and say in one sentence why this series has no endpoints to test. **(c)** Evaluate $R(3)$ exactly.

Answer: ____________________

8. A recursive filter's stored energy is modeled by

$$E(x) = \sum_{n=1}^{\infty} \frac{n\,(x-1)^n}{3^n},$$

where x is the feedback coefficient (dimensionless). **(a)** Apply the ratio test and compute the numerical limit L that remains after factoring out $|x-1|$; state R. **(b)** Test both endpoints. Show what the terms do as $n \to \infty$ and name the test that settles each one. **(c)** Write the complete interval of convergence and justify, in complete sentences, the bracket you gave each endpoint.

Answer: ________________